Arunabha Majumder



Mechanical Engineer, Robotics

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PROFILE

Mechanical engineer in robotics who designs mechanical systems, mechanisms and electromechanical subsystems for robots that work safely with people. I take products from concept and CAD (SolidWorks) to prototypes, integration, validation and testing, with stiffness testing, design optimisation and Design for Manufacturing (DFM) and Design for Assembly (DFA) in mind.

EXPERIENCE

PhD Researcher, Robotics (project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism with a worm gear for pretension, working in parallel with a 6 Nm motor. It uses more than 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Created CAD models and technical drawings in SolidWorks; optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force); 3D printed prototypes, then had parts manufactured.
- Integrated the motor, encoder and torque/force sensors into the structure and measured its stiffness on a physical test setup (validation and testing).
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance.
- DFM in practice: replaced a clutch part that kept breaking and was hard to make with a simple bolt-locking clutch (8 locking angles) that was easy to manufacture and reliable.
- Worked closely with control engineering and software colleagues (team of 3 on an Assist-as-Needed controller); implemented an Arduino-based trigger to sync a robot with Qualisys motion capture.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit (Pneumatic Muscles)
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit; designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW (0.3 - 0.78% error, IEEE CONECCT 2022).

Bachelor's Capstone and Student Teams
Siddaganga Institute of Technology, India | *2017 - 2019*

- Designed and built an arm exoskeleton and exo glove: CAD, machining, assembly and testing (TATA Technologies Prayatna Innovation Fund grant; Best Major Project Award).
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

SKILLS

Mechanical design: mechanisms, machine elements (worm gears, springs, clutches), structural components, electromechanical subsystems, materials selection

CAD and documentation: SolidWorks (CSWA), CAD models, technical drawings, tolerances, stiffness analysis in SolidWorks

Manufacturing: DFM, DFA, 3D printing (Bambu Lab, Ultimaker Cura), machining and assembly

Mechatronics and testing: motors, encoders, torque/force sensors, Arduino, MATLAB/Simulink, LabVIEW, Python (basic), Qualisys, prototype validation

Communication: hand sketching and technical illustration (Senior Diploma in Fine Arts)

EDUCATION

PhD, Mechatronics and Robotics, Aalborg University, Denmark | 2023 - present (research complete)

MTech, Mechatronics, AcSIR - CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering, Siddaganga Institute of Technology, India | 2015 - 2019

Senior Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad | 2008, Distinction

LANGUAGES AND REFERENCES

English (C1), Bengali (native), Hindi (spoken)

References: Prof. Shaoping Bai (sbh@mp.aau.dk); Assoc. Prof. Lars Rosgaard Jensen; Assoc. Prof. Anderson de Souza Castelo Oliveira; all Department of Materials and Production, Aalborg University